Project 2

Technical Report

https://github.com/rmoser10/DTSC-2302-Project-2

Introduction:

Productivity is a big deal in the professional aspects of our lives, from educational to professional; being productive is key. I saw the dataset, which focused on everyday factors that affect focus and productivity, which are two topics I find very interesting. As a college student, I find myself drifting off and scrolling on my phone for an extended period of time instead of just the 5-minute break I wanted to take. This isn't just an issue for students, but rather for most people with social media. There are various metrics used for determining the focus and productivity of someone, and by combining these variables, it can be predicted that they will be a high or low performer. That led me to create this project, which aims to answer the question: Can we predict productivity levels and classify high vs low performers from everyday data?

Data:

The data for this project was found on KAGGLE with the name "Social Media Addiction vs Productivity Dataset". It contains 6000 rows of data with 9 columns, totaling 54000 individual data points. These points are made up mostly of float variables for the metrics of focus, productivity, and daily screen time. The full list of variables with descriptions listed in the dataset is attached below:

1. age: Age of the individual
2. daily_screen_time: Total screen time per day (hours)
3. social_media_hours: Time spent on social media (hours)
4. study_hours: Productive work/study time
5. sleep_hours: Daily sleep duration
6. notifications_per_day: Number of notifications received
7. focus_score: Calculated mental focus (0–100)
8. addiction_level: Low / Medium / High
9. productivity_score: Target variable (0–100)

From these variables, I chose to make productivity_score the target variable, and for my third model, I also included addiction level for classification.

Preprocessing and Exploratory Data Analysis:

At the start of my workflow, I wanted to get an idea of the code, so I chose to use df.head and df.describe statements. Those gave me a good initial idea of what I was dealing with for values and the spreads of those values. In the output for df.describe I was able to discover that each column had only 5880 values, which means each column had 120 values missing. When totaled, those null values account for 2% of the total data points. This was an issue, so I decided to solve it

by utilizing a dropna method to remove the rows with null variables. I then used a heatmap (Figure 1) to explore the correlations between variables, which gave me a better understanding of the connections between variables before I started modeling. Two of the main connections I saw were the productivity score's positive relationship with the study hours and focus score, and its negative correlation with daily screen time and social media hours. After that, I utilized one-hot encoding for the addiction_level variable to be able to work with it in later models.

Modeling:

For my models, I chose to make two decision tree regressors and one regression classifier model. The primary intent of these models was to analyze productivity. The dataset was split 80/20 with random_state=42 to ensure reliable evaluations. The baseline tree utilized default hyperparameters, while the second tree was formed as a tuned decision tree with a max_depth=5, min_samples_split of 10, and min_samples_leaf = 5. I then utilized a logistic regression classifier for the prediction of high and low performance. These combined to give me a good variety of information to form a conclusion with, along with allowing me to determine if the dataset was better at predicting the exact productivity score or classifying an individual as a high or low performer.

Results:

The models all performed well for their intended tasks; the baseline decision tree achieved an R^2 value of 0.7722 (Figure 2). Tree analysis revealed focus score as the primary decision point, followed by study hours. This showed that concentration and time spent studying are the two strongest predictors of productivity. The scatter plot (Figure 4) shows that the predicted values cluster strongly along the identity line, indicating a strong connection between the predictions and the actual values.

The tuned decision tree achieved an R^2 value of only 0.7065, which was ~6.5 percent lower than the baseline. The regularization constraints were overly restrictive, which limited the model to just using 3 variables for its predictions (Figure 3). This demonstrated that hyperparameter tuning could remove useful variables and harm predictability. In contrast to the second model, the logistic regression classifier succeeded at the task of classifying high vs low performers with 90.3% accuracy (Figure 5). The feature coefficients (Figure 6) show that focus_score is the best predictor, with social media hours as a close second. While regression and classification metrics are not directly comparable, both forms identify focus score as crucial, along with general behavioral data being useful for predictions. Model 1 is the ideal choice for exact productivity predictions, and Model 3 is the most valuable for predicting performance as either high or low.

Resources:

https://www.kaggle.com/datasets/asifxzaman/social-media-addiction-vs-productivity-dataset

Python Libraries Used:

- pandas

- **numpy**
- **matplotlib**
- **seaborn**
- **scikit-learn**

AI Acknowledgement:

AI was utilized in the making of this workflow. I utilized GitHub Copilot to help with syntax errors. I also utilized Claude AI to help with fixing coding errors, as well as past scikit demo workflows from DTSC 2 to help form my models.

Appendix:

Figure1:

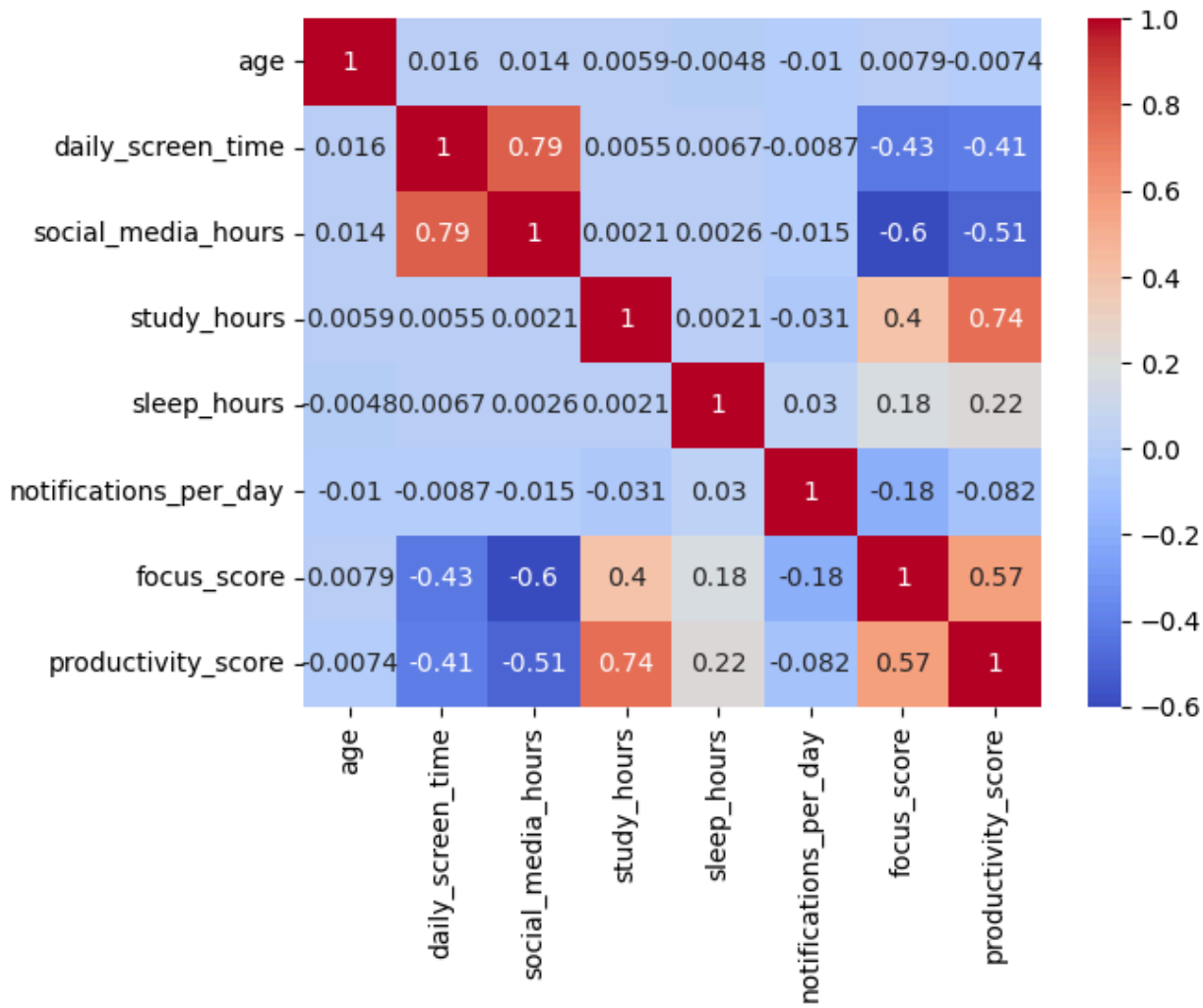


Figure 2:

Model 1: Decision Tree Regressor Tree Structure (max_depth=3 view)

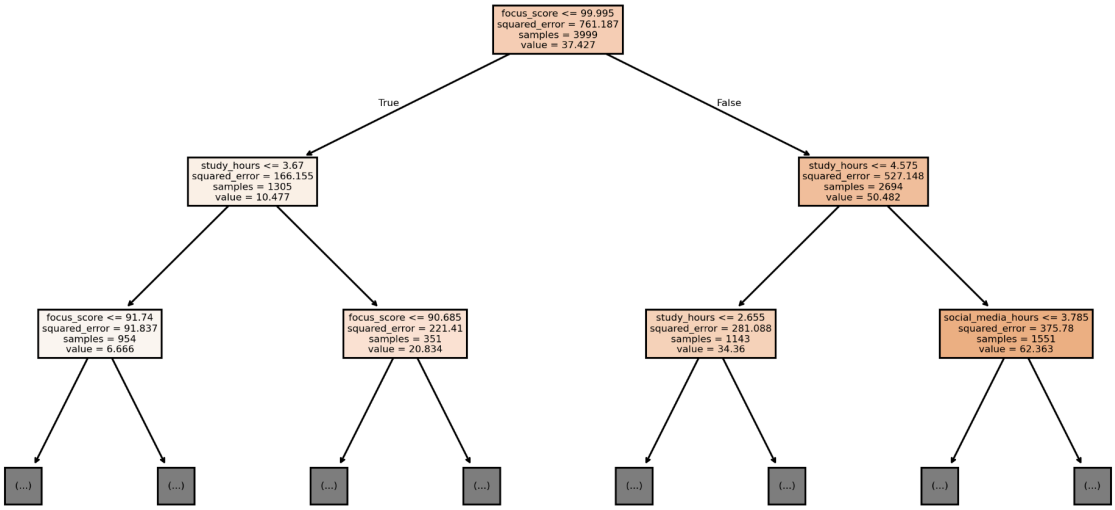


Figure 3:

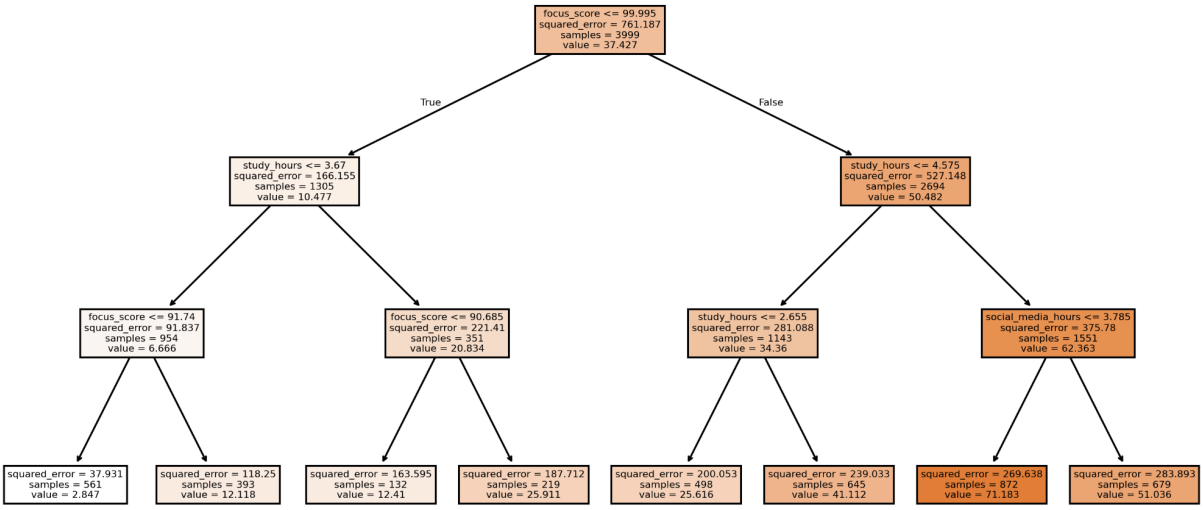
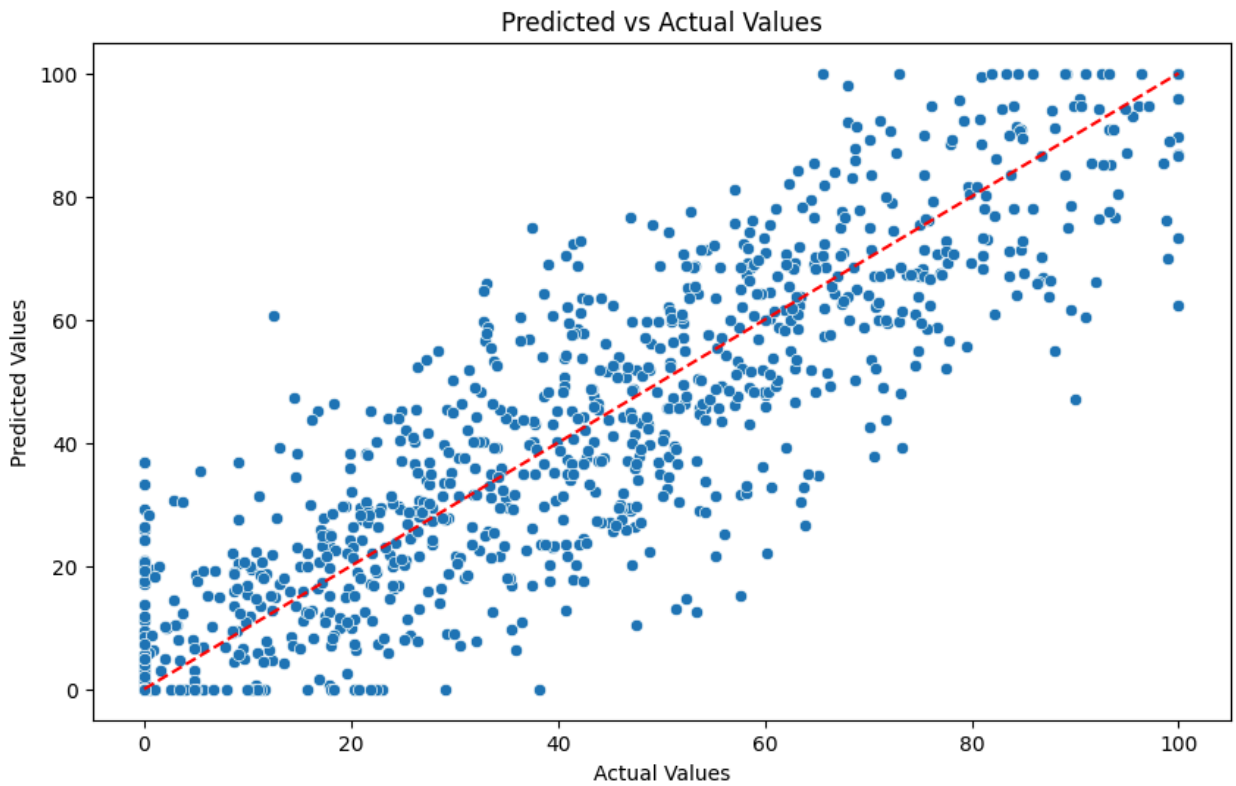
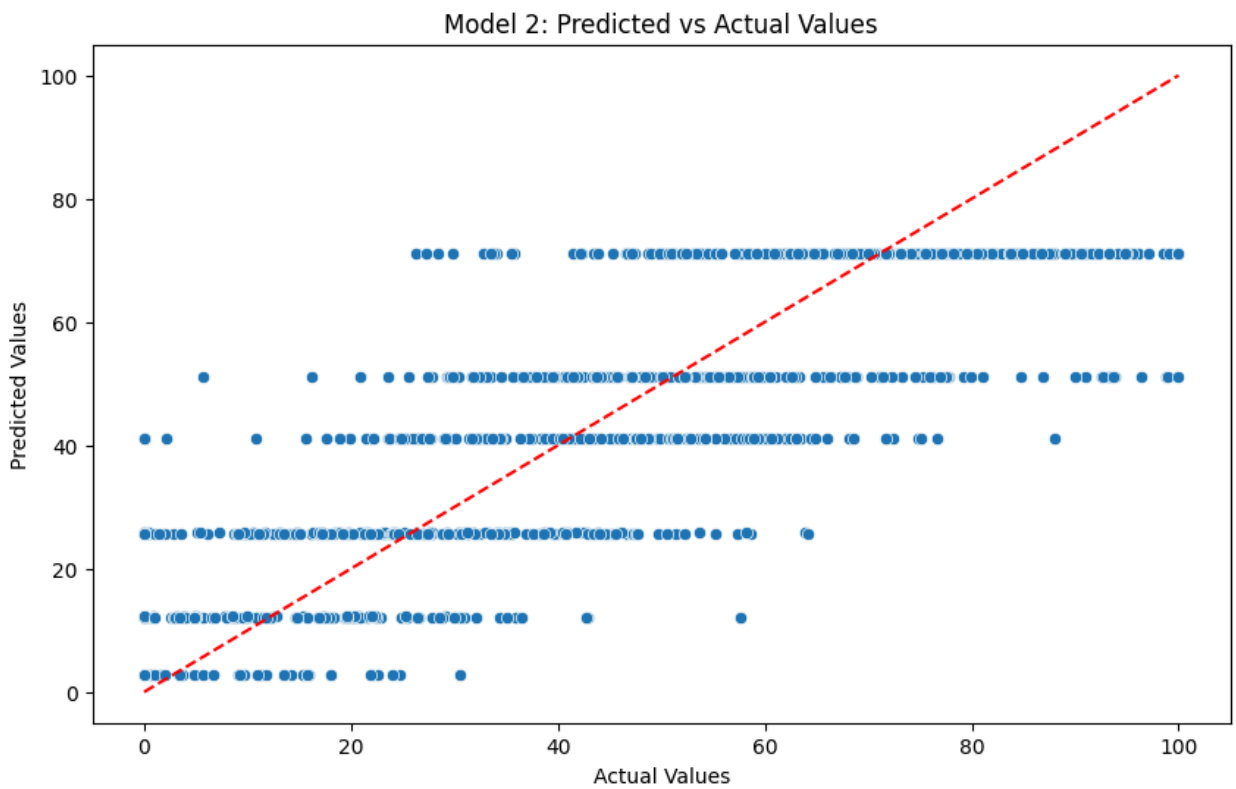


Figure 4:


Predicted vs Actual Values
Predicted Values
Actual Values
0
20
40
60
80
100


Model 2: Predicted vs Actual Values
Predicted Values
Actual Values
0
20
40
60
80
100

	feature	importance
6	focus_score	0.478514
3	study_hours	0.282768
2	social_media_hours	0.121548
4	sleep_hours	0.059385
5	notifications_per_day	0.021531
1	daily_screen_time	0.017491
0	age	0.014695
7	addiction_level_High	0.003851
8	addiction_level_Low	0.000164
9	addiction_level_Medium	0.000051

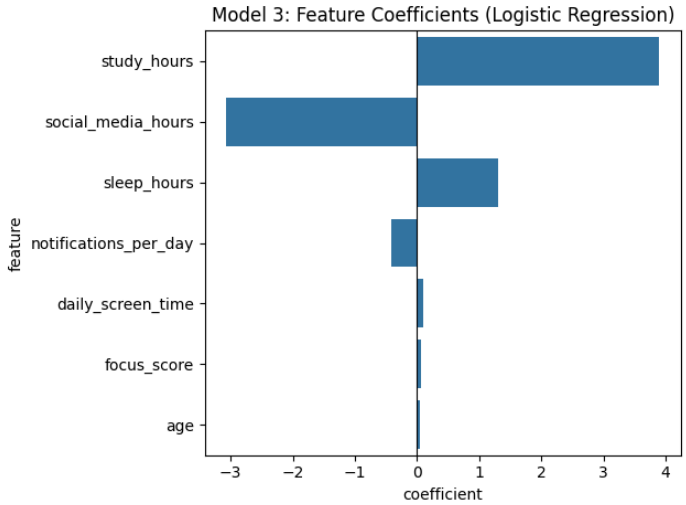
	feature	importance
6	focus_score	0.646053
3	study_hours	0.284521
2	social_media_hours	0.069426
0	age	0.000000
1	daily_screen_time	0.000000
4	sleep_hours	0.000000
5	notifications_per_day	0.000000
7	addiction_level_High	0.000000
8	addiction_level_Low	0.000000
9	addiction_level_Medium	0.000000

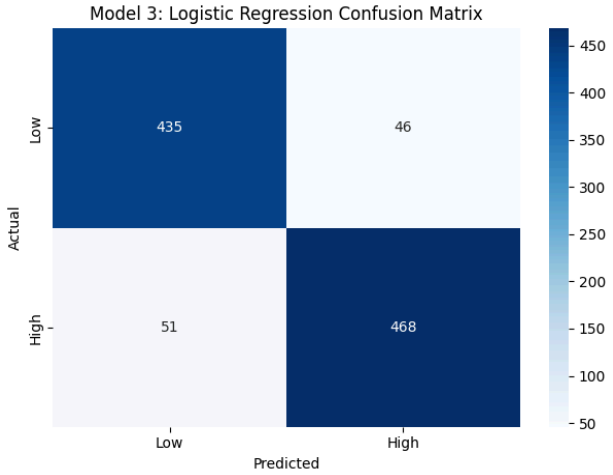
Figure 5:

	Model	Task	Metric	Score
0	Model 1: Baseline DT Regressor	Regression	R^2	0.7682
1	Model 2: Tuned DT Regressor	Regression	R^2	0.7065
2	Model 3: Logistic Regression Classifier	Classification	Accuracy	0.9030

	age	daily_screen_time	social_media_hours	study_hours	sleep_hours	notifications_per_day	focus_score	addiction_level	productivity_score
count	5880.000000	5880.000000	5880.000000	5880.000000	5880.000000	5880.000000	5880.000000	5880	5880.000000
unique	NaN	NaN	NaN	NaN	NaN	NaN	NaN	3	NaN
top	NaN	NaN	NaN	NaN	NaN	NaN	NaN	Medium	NaN
freq	NaN	NaN	NaN	NaN	NaN	NaN	NaN	3064	NaN
mean	27.111395	6.923175	4.145915	4.069328	6.514270	159.781463	96.363531	NaN	37.614051
std	7.268430	2.859055	2.155496	2.309472	1.434351	81.044666	7.416839	NaN	27.603251
min	15.000000	2.000000	0.660000	0.000000	4.000000	20.000000	47.390000	NaN	0.000000
25%	21.000000	4.470000	2.400000	2.120000	5.270000	89.000000	96.360000	NaN	13.565000
50%	27.000000	6.860000	3.755000	4.115000	6.525000	160.000000	100.000000	NaN	36.200000
75%	33.000000	9.360000	5.540000	6.030000	7.730000	231.000000	100.000000	NaN	58.560000
max	39.000000	11.990000	10.660000	8.000000	9.000000	299.000000	100.000000	NaN	100.000000

Figure 6:


Model 3: Feature Coefficients (Logistic Regression)
study_hours
social_media_hours
sleep_hours
notifications_per_day
daily_screen_time
focus_score
age
feature
−3
−2
−1
0
1
2
3
4
coefficient


Model 3: Logistic Regression Confusion Matrix
Low
High
Actual
435
46
51
468
Low
High
Predicted
450
400
350
300
250
200
150
100
50